

Abdul Rafy

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SUMMARY

Backend dev with specializing in Java and Spring Boot. I build REST APIs, third-party integrations, and real-time applications backend systems and like Contributing to open-source.

PROJECTS

PitWall – F1 Live Intelligence Platform

Java, Spring Boot, React, WebSocket, Groq AI

Streamed live Formula 1 telemetry by integrating the OpenF1 API with a 30-second polling interval, broadcasting driver positions and timing data to a React frontend via WebSocket. Integrated Groq LLM to generate contextual race strategy insights during live sessions and aggregated official F1 highlights from YouTube in real time. Deployed frontend on Vercel and backend on Render – [Live Demo Link](#).

Apex – Real-Time Market Simulation

Java, Spring Boot, PostgreSQL, Redis, RabbitMQ, React

An enterprise-scale paper trading platform with an idempotent, optimistically-locked trading engine preventing double-spends under concurrent trades. Implemented a WebSocket gateway with Redis pub/sub fan-out and an analytics engine computing Sharpe ratio and drawdown from trade history. Designed multi-tenant RBAC with org leaderboards and an AI-generated daily trade journal via Claude/Gemini.

Argent – Financial Infrastructure

Java, Spring Boot, PostgreSQL, Redis, React, TypeScript

Built a multi-tenant wallet and double-entry ledger engine with database-enforced immutability and optimistic locking against concurrent balance races, exposing sandbox/production-isolated REST APIs secured via JWT and API keys with full audit logging. Implemented deposit, transfer, and refund engines via strategy pattern, validated by 160+ automated tests, with a React dashboard for wallet management.

Anvil – Distributed Job Processing

Java, Spring Boot, Spring Security, JPA, PostgreSQL, Redis

Built a job orchestration platform with JWT auth, RBAC, and a centralized state machine for job lifecycle transitions. Used a strategy-pattern handler to decouple job logic from the queue/worker layer, with Flyway migrations. Implemented exponential-backoff retries and a dead-letter queue.

EXPERIENCE

Backend Engineering Trainee – FlyRank AI

Remote

Selected for FlyRank AI's Backend AI Engineering track, focused on AI-powered applications, backend development, API engineering, and modern software engineering practices.

EDUCATION

2023 – 2027 BS in Computer Science, **FAST National University**, Karachi

GPA: 3.38 / 4.0

SKILLS

Programming Languages Java, Python, C#, C++

Backend & Web Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, WebSocket, React, TypeScript

Databases PostgreSQL, MySQL, MongoDB, SQLite, pgvector

Messaging & Caching RabbitMQ, Redis

Tools & Build Git, GitHub, Docker, Docker Compose, Maven, IntelliJ IDEA

ACHIEVEMENTS

- Dean's List, FAST National University (SGPA 3.5+) – 2 semesters.
- Teaching Assistant, OOP, FAST NUCES – assisted ~50 students with debugging, grading, and sessions.